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Experiment 04 : Installation of Java, Jenkins and CI Pipeline using Jenkins

Based on the uploaded report and screenshots. This document expands each step in detail while preserving the workflow shown in the uploaded PDF.

Objective
1. Install Java Development Kit (JDK).
2. Verify Java installation using command prompt.
3. Download and configure Jenkins using the Generic Java Package (.war).
4. Install suggested Jenkins plugins and create the first administrator account.
5. Create a Jenkins Pipeline for a Java application.
6. Demonstrate Continuous Integration by automatically compiling and executing a Java program.
7. Modify the Java program and rebuild the pipeline to verify that Jenkins automatically uses the latest source code.

Software Requirements
• Windows 11
• Oracle JDK 21.0.11 (64-bit)
• Jenkins 2.568.1 LTS (.war)
• Command Prompt
• Web Browser
• Internet Connection

Procedure
A. Java Installation
• Verified that Java was not installed using:
  - where java
  - java -version
• Installed Oracle JDK 21.0.11.
• Verified installation using:
  - java -version
  - javac -version

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- where java
- echo %JAVA_HOME%

Expected result: Java compiler and runtime available.

B. Jenkins Installation

• Downloaded Jenkins 2.568.1 LTS (Generic Java Package - jenkins.war).
• Created C:\Jenkins.
• Moved jenkins.war into C:\Jenkins.
• Started Jenkins using:
  java -jar jenkins.war
• Opened http://localhost:8080.
• Unlocked Jenkins using the generated initial administrator password.
• Selected 'Install suggested plugins'.
• Waited for all plugins to install successfully.
• Created the first administrator user.
• Completed instance configuration.
• Opened the Jenkins Dashboard.

C. Creating the CI Pipeline

• Created project folder C:\HelloWorldProject.
• Created HelloWorld.java.
• Verified manual execution using:
  javac HelloWorld.java
  java HelloWorld
• Created a new Jenkins Pipeline named HelloWorldPipeline.
• Selected 'Pipeline Script' as the definition.
• Added a pipeline with two stages:
  Stage 1 – Compile Java
  Stage 2 – Run Java
• Saved the pipeline.
• Clicked 'Build Now'.
• Verified successful execution from Console Output.

D. Continuous Integration Demonstration

• Modified HelloWorld.java to print enhanced build information.
• Ran 'Build Now' again.
• Jenkins automatically compiled and executed the updated code without changing the pipeline configuration, demonstrating Continuous Integration.

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Pipeline Script Used

```
pipeline {
    agent any
    stages {
        stage('Compile Java') {
            steps {
                dir('C:/HelloWorldProject') {
                    bat 'javac HelloWorld.java'
                }
            }
        }
        stage('Run Java') {
            steps {
                dir('C:/HelloWorldProject') {
                    bat 'java HelloWorld'
                }
            }
        }
    }
}
```

Explanation of CI Workflow

Developer updates HelloWorld.java
↓
Jenkins Pipeline is triggered
↓
Compile source code (javac)
↓
Execute application (java)
↓
Display console output
↓
Report SUCCESS or FAILURE

The final build in the uploaded report shows that after modifying the Java source code, Jenkins executed the updated program successfully without changing the pipeline itself, which demonstrates the concept of Continuous Integration.

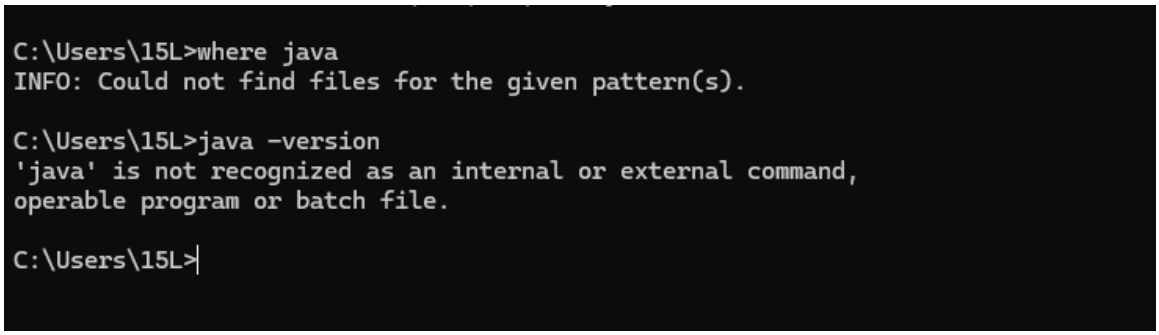
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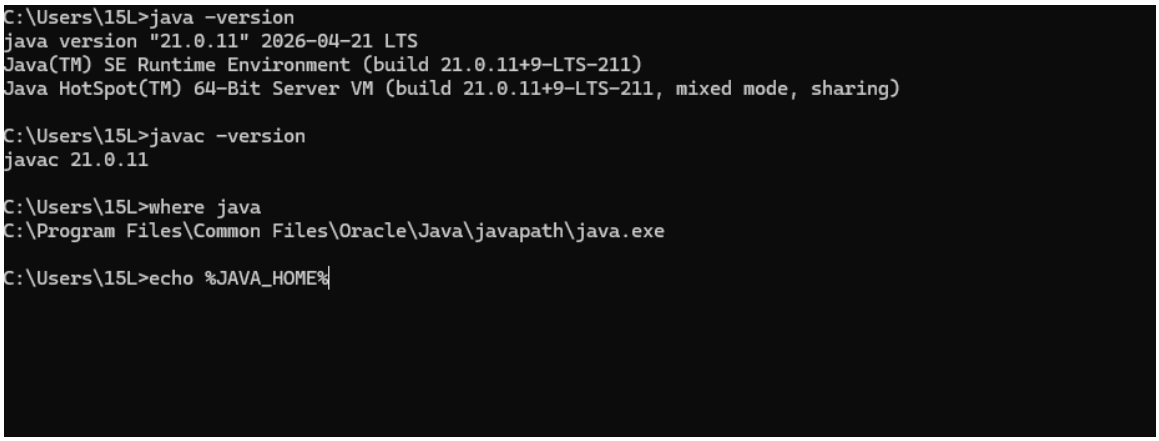
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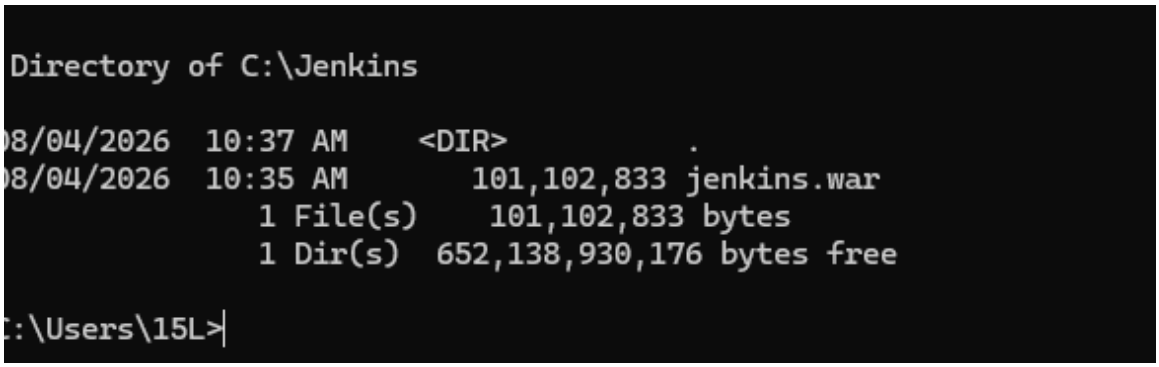
Screenshots Description

Page 1: Verification that Java was initially absent followed by successful JDK installation.



Page 2: Verification commands confirming Java installation and Jenkins WAR placed in C:\Jenkins.



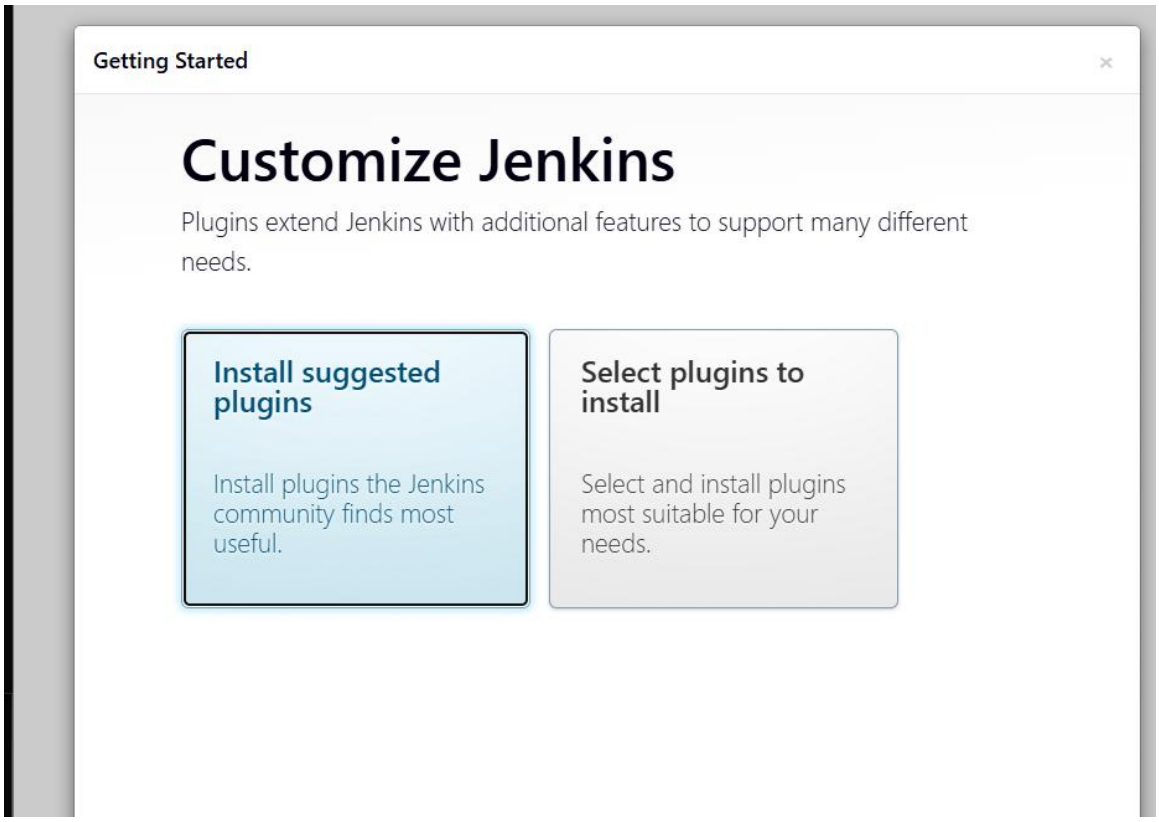


Page 3: Jenkins launched successfully and Customize Jenkins page displayed.

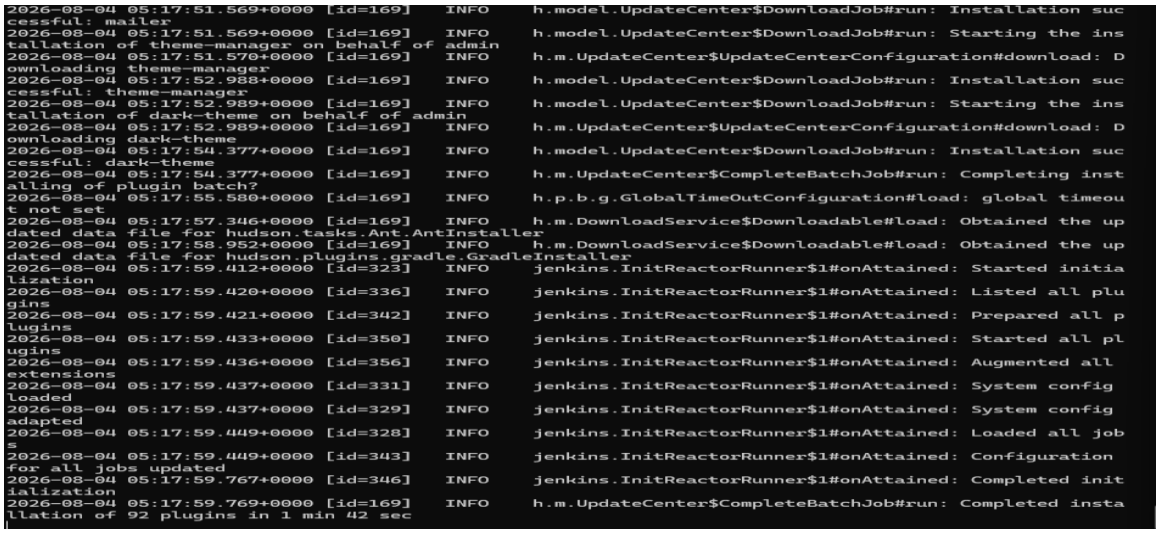
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Page 4: Installation of suggested plugins completed successfully.

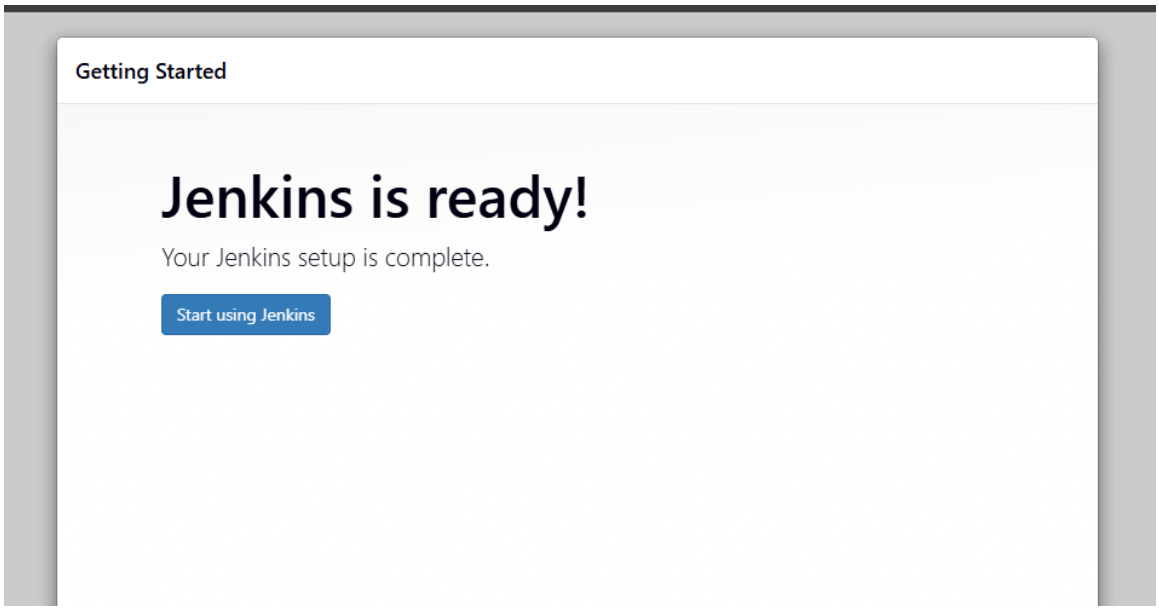


Page 5: Jenkins setup completed and ready to use.

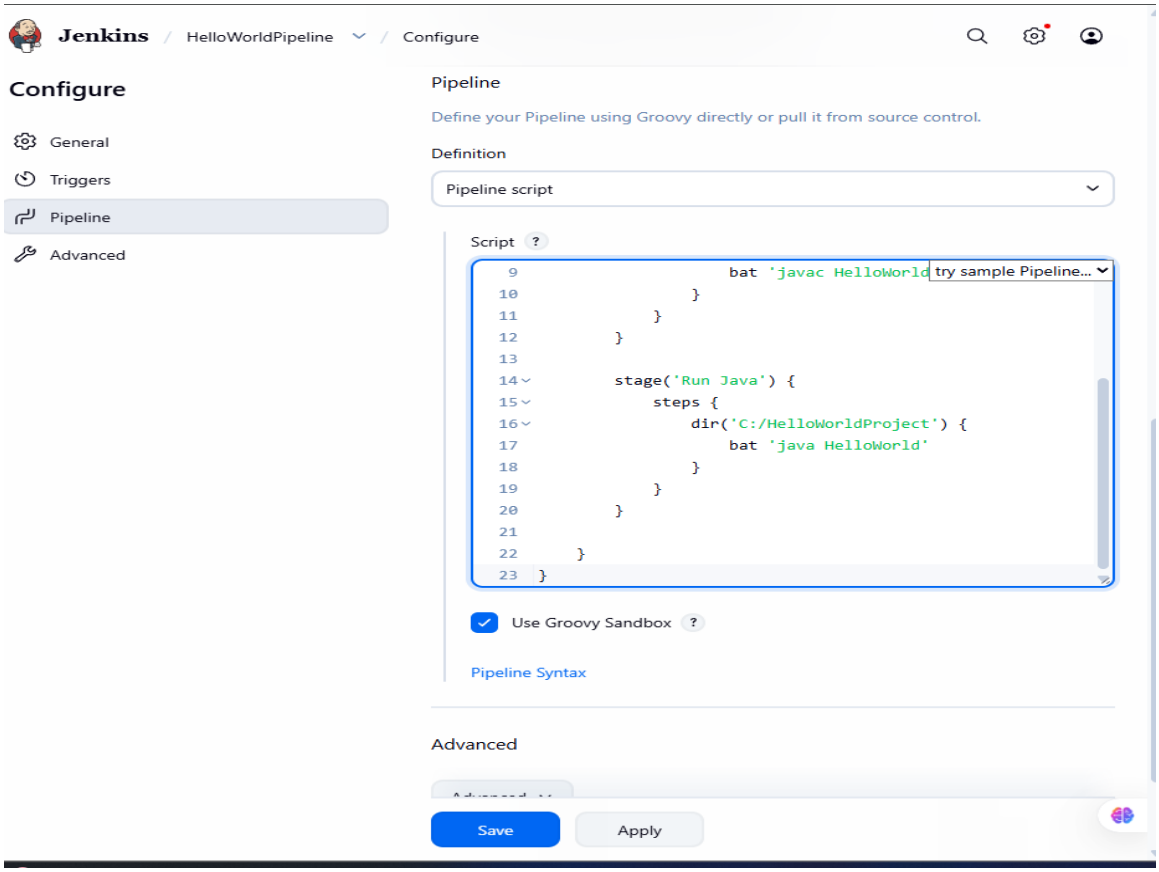
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Page 6: Creation of HelloWorldPipeline using a Pipeline Script.



Page 7: First successful pipeline execution showing compilation and execution.

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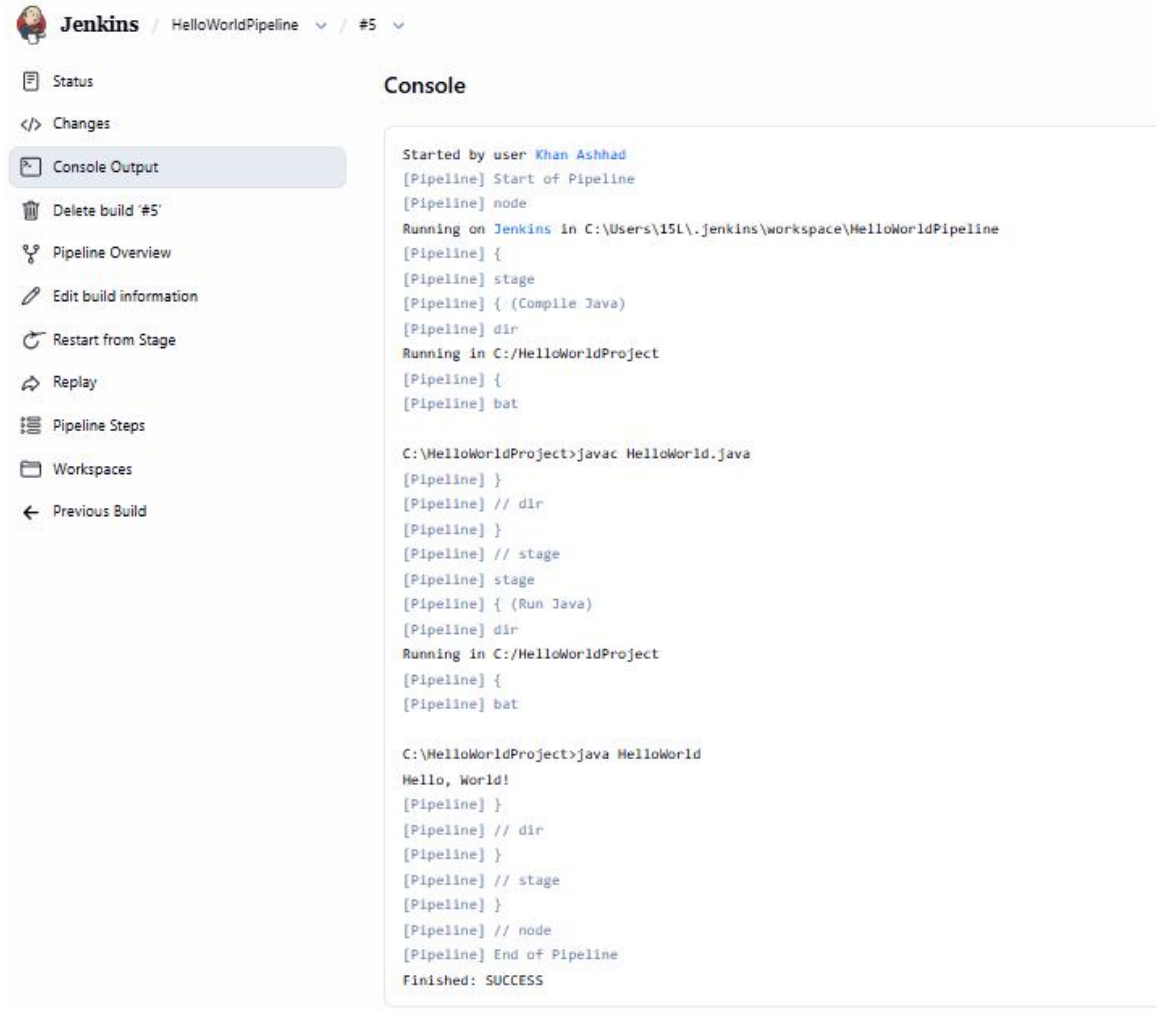
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Page 8: Updated Java source code rebuilt successfully, demonstrating CI.

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Console

```
Started by user Khan Ashhad
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\HelloWorldPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Compile Java)
[Pipeline] dir
Running in C:/HelloWorldProject
[Pipeline] {
[Pipeline] bat

C:\HelloWorldProject>javac HelloWorld.java
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Run Java)
[Pipeline] dir
Running in C:/HelloWorldProject
[Pipeline] {
[Pipeline] bat

C:\HelloWorldProject>java HelloWorld
==========================================
      JENKINS CONTINUOUS INTEGRATION
==========================================
Project      : HelloWorld CI
Java Version : 21.0.11
OS           : Windows 11
Build Time   : 04-08-2026 11:14:14

[PASS] Source code compilation successful.
[PASS] Unit tests executed successfully.
[PASS] Build packaged successfully.

BUILD STATUS : SUCCESS
READY FOR DEPLOYMENT
==========================================
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Result

Java JDK and Jenkins were successfully installed and configured. A Jenkins Pipeline was created to automatically compile and execute a Java program. After modifying the source code, the pipeline was executed again and Jenkins successfully built and ran the latest version of the application, demonstrating the working of Continuous Integration.

Conclusion

This experiment demonstrated the complete setup of a Jenkins-based Continuous Integration environment. Java was installed and verified, Jenkins was configured from the

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Generic Java Package, required plugins were installed, and a pipeline was created to automate the compilation and execution of a Java program. Rebuilding the pipeline after modifying the source code confirmed that Jenkins automatically integrates and validates the latest changes, illustrating the fundamental concept of Continuous Integration used in modern DevOps practices.